

Data Analytics for Business

Homework 1 – Part 2

Linear Models (2)

In this homework, we will fit non-linear patterns in data as well as include qualitative variables and interaction terms into the model. We will work on the same data set of used car sales records as in Part 1. This time, we add a qualitative/categorical variable, **Color**, into the data. The data fields are described below.

Data Description

Column Name	Variable Description
Id	ID number of each used car
Model	Model name of each used car
Price	The price (in Euros) at which each used car was sold
Age	Age (in months) of each used car as of August 2004
KM	Accumulated kilometers on odometer
HP	Horsepower
Metallic	Metallic color? (Yes = 1, No = 0)
Automatic	Automatic transmission? (Yes = 1, No = 0)
CC	Cylinder volume (in cubic centimeters)
Doors	Number of doors
Gears	Number of gears
Weight	Weight (in kilograms)
Color	The color of a car

Download the data file "[**UsedCars2.csv**](#)". Complete the tasks outlined below.

Tasks:

1. Import "UsedCars2.csv." Set the **stringsAsFactors** option to **TRUE** so the string variables are treated as **factors**. Examine the structure of the data to make sure categorical variables are properly stored as factors.
2. According to our analysis from last homework, we already find those variables that have significant effects on Price. Run a linear regression of **Price** on these variables (i.e., **Age, KM, HP, Automatic, Gears, Weight**) and also include **Color**. Show the regression result summary. Discuss how Color is treated in the regression. How many dummy variables are created, and which color is the baseline? Use **contrasts()** to show how the dummy variables are coded. What does each dummy variable's coefficient mean? Does the color of a car have any significant effect on the sales price?
3. Following Step 2, exclude **Color** from the regression, but include an **interaction term** between **Age** and **KM** in the regression. (Notice that you also need to include the **main effects** of Age and KM.) Show the regression result summary and discuss the implications. Is the interaction effect between Age and KM significant? If so, what does it mean?
4. Create a scatterplot of **KM** and **Price**. Is there any non-linear relationship? If so, briefly describe such a non-linear relationship in words. We can capture such a non-linear relationship in a regression along with other independent variables. This is called a **Generalized Additive Model (GAM)**. Regress **Price** on **Polynomials** of **KM** up to **degree 4** along with another variable **Automatic**. Set "**raw=TRUE**" to use the standard basis functions for the polynomials. Show the regression result summary and see how many coefficients there are in total. Add the curve of fitted **Price** versus **KM** onto the plot. Notice that you need to specify the value of **Automatic** (as the mean value from the data) in order to produce the curve.

Hint: You can use the following R code to plot the curve (where **lm3** is the Polynomial regression result).

```
km.grid <- seq(from=min(KM), to=max(KM), by=1000)
preds <- predict(lm3, newdata=list(KM=km.grid, Automatic=rep(mean(Automatic),length(km.grid))))
```